

Knapsack

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Practice quiz

Patricia is about to spend several days on a long hike in the mountain to climb the Bishorn. She is now preparing her knapsack and thinking about which items to take. The alpine guides strongly recommend carrying no more than 7kg. Therefore, Patricia has to decide which items to carry and which items to leave at home. Each of them has a different level of importance. While it is critical to carry water and food it is less critical to carry a laptop. For each item i considered by Patricia, she knows its weight in kilos $w_i \geq 0$ and its value $u_i \in \mathbb{R}$.

Item	Value	Weight
Laptop	1	1
Water	4	3
Food	5	4
Clothes	7	5

1. Write the mathematical model for this integer optimization problem.
2. Determine which items Patricia should take in order to maximize the total utility while satisfying the maximum weight constraints.